DATA ITEM DESCRIPTION
Title: SURVIVABILITY COST EFFECTIVENESS TRADEOFF STUDIES REPORT

Number: DI-SESS-81500A
AMSC Number: N10105
DTIC Applicable: N/A
Preparing Activity: AS
Applicable Forms: N/A

Approval Date: 20191008
Limitation: N/A
GIDEP Applicable: N/A
Project Number: SESS-2019-055

Use/Relationship: This report presents the results of three interdependent analyses. These are used to indicate the probability that aircraft will survive specified combat missions against specified threats and to provide measures of cost effectiveness and design tradeoff studies by which candidate survivability enhancement techniques may be evaluated. These studies are designed to support cost and operational effectiveness studies which help define basic operational and effectiveness in detailed mission scenarios.

The primary purpose of this report is to support the assessment of the operational effectiveness and suitability of systems design concepts and characteristics in threat environments. The results of these studies may be used in comparting alternative systems or force mixes such as in cost and operational effectiveness assessments and to assist in evolutionary requirements definition and in the systems engineering process. They are indispensable in comparing alternative concepts in the pre-Milestone I timeframe and in operational requirements. This report should be acquired whenever an evaluation of the survivability of a system is required. This report applies to both development and non-developmental items.

This Data Item Description (DID) contains the format, content, and intended use information for the data deliverable resulting from the work task described in the contract.

Requirements:

1. Referenced Documents: None

2. Format: Contractor format is acceptable.

3. Content: The report shall consist of two sections:

3.1 Survivability Assessment Section. This section shall include data that permits determination of the effectiveness of the system and all existing and proposed survival enhancement techniques under a variety of encounter conditions and also provide quantified levels of system survivability.

3.1.1 This section shall contain a description of the threat and encounter conditions, a description of all the survivability enhancement techniques considered, identification of the source of all data used in the calculations, description of the methodology used, and results.

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3.1.2 The report shall describe the effects of observables, threat warning and avoidance capabilities, active and passive countermeasures (both on-board and off-board), electronic warfare techniques, maneuvering, speed, flight altitude, threat tolerance (i.e., vulnerability reduction realized with respect to specified threats and kill criteria), day/night and weather, and armament performance, tactics, and doctrine shall be determined. Flight profiles and tactics used by the air vehicle shall be identified to include: (1) base escape (departure); (2) ingress and egress speed and altitude; (3) threat avoidance maneuvers; (4) use of countermeasures, and (5) target area profiles including target acquisition, re-acquisition, weapon delivery, and post-delivery. The report shall recommend the testing required verifying the achievement of the assessed survivability related operational effectiveness and suitability characteristics. The analysis shall are based on applicable results of the mission-threat encounter and vulnerability analyses.

3.2 Survivability enhancement tradeoff studies section. This section shall identify the effects of variations in each significant survivability analysis parameter (e.g. threat, operational utilization, performance, system baseline characteristics, and incorporation of survivability enhancement techniques) on overall operational effectiveness and suitability, acquisition cost and schedule, and life cycle costs. This section describes the results of cost and operational effectiveness trade studies of the variable listed below:

3.2.1 All of the parameters described in 3.1 2 above.

3.2.2 Impacts on weight performance, cost, reliability, maintainability, safety, ease of reparability, and producibility.

3.2.3 Recommendations and alternatives regarding optimum design configuration and other survivability enhancement techniques or doctrine.

3.2.4 Details concerning installation and removal, if “kit form” techniques are recommended for use.

3.3 Identification of the following shall be included:

3.3.1 Level (s) of analysis to be performed (i.e., one-on-one, few-on-few, campaign, base escape).

3.3.2 Missions, targets/target arrays, weapons carried, delivery profiles, and threats/threat arrays to be analyzed.

3.3.3 Analysis methods and models to be used.

3.3.4 Support systems (e.g., AWACS, Wild Weasel, EA-6B, etc.) to be considered.

3.3.5 Scope of effort for campaign analyses (e.g. single mission blue air versus red defenses over time, fully two-sided threater level etc.). A multiple scope analysis can be specified as discrete tasks to provide visibility into the key issues being investigated.

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3.3.6 Specific trade area (s) to be addressed.

3.3.7 Specific measures of effectiveness to be used.

3.3.8 Analysis methods or models to be used.

End of DI-SESS-81500A

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